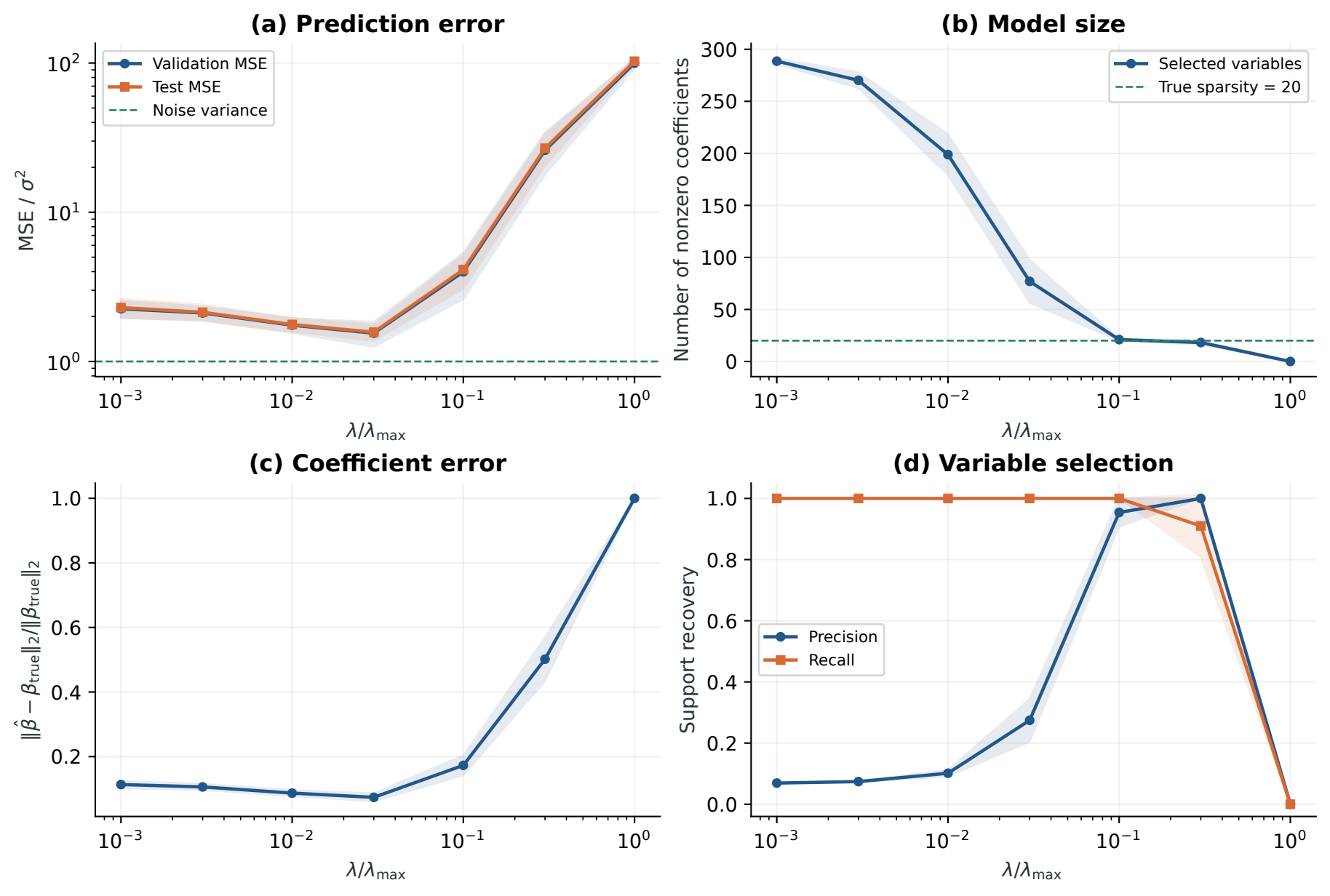
(a) Prediction error
Validation MSE
Test MSE
Noise variance
MSE / σ^2
10^2
10^1
10^0
10^{-3}
10^{-2}
10^{-1}
10^0
$\lambda/\lambda_{\max}$
(b) Model size
Selected variables
True sparsity = 20
Number of nonzero coefficients
300
250
200
150
100
50
0
10^{-3}
10^{-2}
10^{-1}
10^0
$\lambda/\lambda_{\max}$
(c) Coefficient error
$\|\hat{\beta} - \beta_{\text{true}}\|_2 / \|\beta_{\text{true}}\|_2$
1.0
0.8
0.6
0.4
0.2
10^{-3}
10^{-2}
10^{-1}
10^0
$\lambda/\lambda_{\max}$
(d) Variable selection
Precision
Recall
Support recovery
1.0
0.8
0.6
0.4
0.2
0.0
10^{-3}
10^{-2}
10^{-1}
10^0
$\lambda/\lambda_{\max}$